Make Geometry Matter for Spatial Reasoning

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Research Questions

- Why do geometry tokens injected from 3D foundation models often remain underutilized in VLMs for spatial reasoning despite their potential?
- How can we force VLMs to actively consult geometry tokens rather than relying on 2D visual shortcuts for spatial reasoning?
- What mechanisms can adaptively route geometry information to regions where geometric evidence is most critical?

Core Idea

The paper addresses a counterintuitive finding: when geometry tokens from 3D models are naively injected into vision-language models, they often get ignored in favor of 2D visual shortcuts, sometimes even hurting performance. The key insight is that models need to be forced to use geometric information through strategic masking of 2D tokens during training and adaptive fusion that amplifies geometry contributions only where needed. This works because it removes the easy path (2D shortcuts) while providing fine-grained control over when geometric evidence matters most.

Key Formula

Geometry-Unleashing Masking

```
\tilde{\boldsymbol{F}}_\text{V} = \boldsymbol{m} \odot \boldsymbol{F}_\text{V}, \quad m_j = \begin{cases} 0, & \text{if } \delta = 1 \wedge j \in \mathcal{M} \\ 1, & \text{otherwise} \end{cases}, \quad \delta \sim \text{Bernoulli}(\beta)
```

F_V are vision tokens, m is binary mask, M is mask set, β controls masking probability during training, δ enables/disables masking. This strategically removes 2D visual information to force the model to consult geometry tokens for spatial reasoning.

Geometry-Guided Fusion

$$\boldsymbol{\alpha} = \sigma(\mathbf{W}_g[\boldsymbol{V} \| \boldsymbol{G}] + \boldsymbol{b}_g), \quad \boldsymbol{F} = \boldsymbol{\alpha} \odot \boldsymbol{V} + (1 - \boldsymbol{\alpha}) \odot \boldsymbol{G}$$

V are layer-normalized vision features, G are geometry features, α is learned gating weights, W_g and b_g are learnable parameters. This adaptively controls geometry vs vision contribution at each spatial location and channel, allowing geometry to dominate where it's most informative.

Key Figure

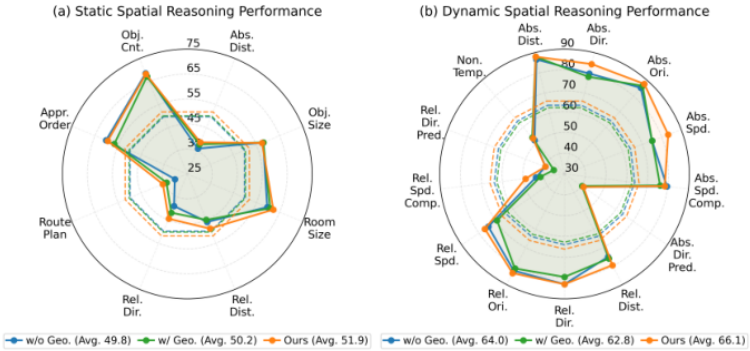

Fig. 1: Geometry injection can be underutilized and even harmful for spatial reasoning. We compare three variants: w/o Geo. (removing the geometry branch), w/ Geo. (injecting geometry tokens via naive fusion with standard fine-tuning), and Ours. The plots reveal a counterintuitive yet reproducible pattern: in static scenes, w/ Geo. brings only marginal gains over w/o Geo., whereas in dynamic videos, w/ Geo. can even underperform the w/o Geo. baseline. This suggests that VLMs often fall back to appearance-driven shortcuts in 2D visual tokens and treat geometry as a dispensable side signal. These observations motivate **GeoSR** (Ours), which compels models to use geometry as actionable evidence and yields consistent improvements across both static and dynamic spatial reasoning.

tokens and frames. Together, these designs make geometry tokens actionable and controllable, rather than being easily ignored or indiscriminately fused.

In summary, our main contributions are as follows.

- We report a reproducible finding in spatial reasoning that implicit geometry injection can be ineffective under naive fusion and standard fine-tuning. Geometry tokens may be largely ignored, with limited or even negative gains on spatial reasoning in practice.
- We propose **GeoSR**, a simple framework that encourages effective and reasonable use of geometry tokens. Geometry-Unleashing Masking reduces reliance on appearance shortcuts, and Geometry-Guided Fusion routes geometry information adaptively when geometric evidence is required.
- We validate **GeoSR** on both static and dynamic spatial reasoning benchmarks, showing consistent improvements over prior methods.

2 Related Work

2.1 General Video Understanding

Recent advances have extended vision-language models from static images to videos by representing a clip as multiple sampled frames and modeling temporal

Figure 1: Demonstrates the core counterintuitive finding that naive geometry injection ('w/ Geo.') provides minimal gains in static scenes and can even hurt performance in dynamic scenes compared to no geometry baseline, motivating the need for GeoSR.

Limitations

- Relies on quality of pretrained 3D foundation models (VGGT, π^3) for geometry token extraction
- Masking hyperparameters (γ=0.8, β=0.5) may need tuning for different domains or datasets
- Current datasets may have ambiguous geometric annotations that limit all methods' performance
- Computational overhead from additional geometry branch and gating mechanisms, though reported as minimal